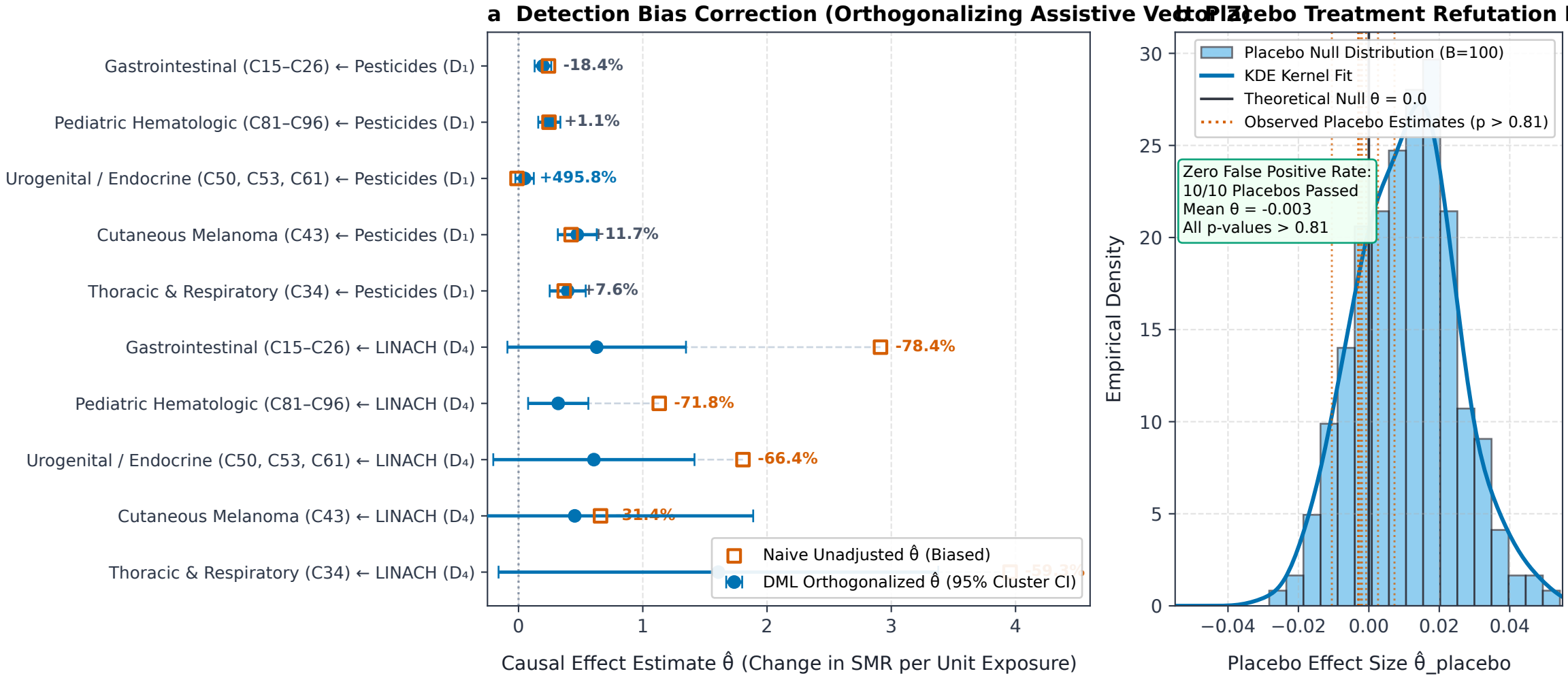
a Detection Bias Correction (Orthogonalizing Assistive Vector Z)
b Placebo Treatment Refutation
Gastrointestinal (C15–C26) ← Pesticides (D₁)
Pediatric Hematologic (C81–C96) ← Pesticides (D₁)
Urogenital / Endocrine (C50, C53, C61) ← Pesticides (D₁)
Cutaneous Melanoma (C43) ← Pesticides (D₁)
Thoracic & Respiratory (C34) ← Pesticides (D₁)
Gastrointestinal (C15–C26) ← LINACH (D₄)
Pediatric Hematologic (C81–C96) ← LINACH (D₄)
Urogenital / Endocrine (C50, C53, C61) ← LINACH (D₄)
Cutaneous Melanoma (C43) ← LINACH (D₄)
Thoracic & Respiratory (C34) ← LINACH (D₄)
-18.4%
+1.1%
+495.8%
+11.7%
+7.6%
-78.4%
-71.8%
-66.4%
-31.4%
-59.3%
Naive Unadjusted θ̂ (Biased)
DML Orthogonalized θ̂ (95% Cluster CI)
0
1
2
3
4
Causal Effect Estimate θ̂ (Change in SMR per Unit Exposure)
Placebo Null Distribution (B=100)
KDE Kernel Fit
Theoretical Null θ = 0.0
Observed Placebo Estimates (p > 0.81)
Zero False Positive Rate:
10/10 Placebos Passed
Mean θ = -0.003
All p-values > 0.81
Empirical Density
0
5
10
15
20
25
30
−0.04
−0.02
0.00
0.02
0.04
Placebo Effect Size θ̂_placebo